

ORIGINAL RESEARCH

Heart Failure Duration, Cardiac Remodeling, Dysfunction, and Hemodynamic Severity in HFpEF and HFmrEF

Insights From REDUCE LAP-HF II

Shunichi Doi, MD, PhD,^a Sho Kazui, MD, PhD,^a Naoyuki Akashi, MD,^a Ravi B. Patel, MD, MSc,^b Jan Komtebedde, DVM,^c Uma Mylavarapu, MPH, MHA,^b Vijay Swarup, MD,^d Sebastian Winkler, MD,^e Sören J. Backhaus, MD,^f Sheldon E. Litwin, MD,^{g,h} Rajeev C. Mohan, MD,ⁱ Rami Kawash, MD,^j Aaron L. Sverdlov, MBBS, PhD,^k Eugene S. Chung, MD,^l David M. Kaye, MD,^{m,n,o,p} Elke Hoendermis, MD, PhD,^q Scott L. Hummel, MD, MS,^r Maja Cikes, MD, PhD,^s Christopher S. Hayward, MD,^t Randall Starling, MD, MPH,^u Deepak K. Gupta, MD, MSCI,^v Andreas Zirlik, MD,^w Scott McKenzie, MD,^x Ajith Nair, MD,^y Andreas Rieth, MD,^z Donald E. Cutlip, MD,^{aa} Scott D. Solomon, MD,^{bb} Dirk J. van Veldhuisen, MD, PhD,^{cc} Martin B. Leon, MD,^{dd} Frank E. Silvestry, MD,^{ee} Sanjiv J. Shah, MD,^b Barry A. Borlaug, MD^a

ABSTRACT

BACKGROUND Pathophysiologic features associated with increased risk in heart failure with preserved ejection fraction (HFpEF) may reflect distinct phenotypes or disease stages related to heart failure (HF) duration.

OBJECTIVES This study aims to determine how duration of HF diagnosis influences cardiac structure and function, hemodynamics, risk for worsening HF, and treatment response.

METHODS Baseline cardiac structure and function, hemodynamics, and outcomes following randomization were evaluated among individuals participating in the REDUCE LAP-HF II trial (N = 626): patients with HF, left ventricular ejection fraction $\geq 40\%$, and exercise pulmonary capillary wedge pressure ≥ 25 mm Hg, categorized by HF duration: <1 year, 1–3 years, and >3 years.

RESULTS Longer HF duration was associated with greater cardiac remodeling and dysfunction, including lower left ventricular global longitudinal strain (16.4% [Q1–Q3: 14.3%–18.6%] vs 17.7% [Q1–Q3: 15.4%–20.0%] vs 18.4% [Q1–Q3: 16.3%–20.5%]; $P < 0.0001$) and right ventricular free wall strain (20.8% [Q1–Q3: 16.7%–24.6%] vs 22.2% [Q1–Q3: 17.6%–26.1%] vs 24.4% [Q1–Q3: 19.2%–27.2%]; $P < 0.0001$), and more biatrial myopathy (48.8% vs 40.8% vs 28.2%; $P < 0.0001$). Resting pulmonary capillary wedge pressure elevation was more common with longer duration (75.8% vs 70.7% vs 64.9%; $P = 0.024$). These findings were confirmed in an independent sample of patients with HFpEF undergoing evaluation at Mayo Clinic. HF event risk increased with longer duration (HR: 1.70 [95% CI: 1.17–2.48]; $P = 0.006$ for >3 years vs <1 year). No significant heterogeneity in atrial shunt response was observed across duration groups.

CONCLUSIONS Longer duration of HFpEF diagnosis is associated with more advanced cardiac remodeling and dysfunction and higher risk of worsening HF, underlining the priority for earlier diagnosis to improve outcomes. (A Study to Evaluate the Corvia Medical, Inc IASD System II to Reduce Elevated Left Atrial Pressure in Patients with Heart Failure [REDUCE LAP-HF TRIAL II]; [NCT03088033](https://doi.org/10.1016/j.jchf.2026.103167)) (JACC Heart Fail. 2026;■:103167) © 2026 by the American College of Cardiology Foundation.

ABBREVIATIONS
AND ACRONYMS

GLS = global longitudinal strain

HF = heart failure

HFmrEF = heart failure with mildly reduced ejection fraction

HFpEF = heart failure with preserved ejection fraction

KCCQ-OSS = Kansas City Cardiomyopathy Questionnaire-Overall Summary Score

LA = left atrium

LV = left ventricle

PA = pulmonary artery

PCWP = pulmonary capillary wedge pressure

PVD = pulmonary vascular disease

PVR = pulmonary vascular resistance

RA = right atrium

RV = right ventricle

TAPSE = tricuspid annular plane systolic excursion

Heart failure with preserved ejection fraction (HFpEF) or heart failure with mildly reduced ejection fraction (HFmrEF) accounts for >50% of all heart failure (HF) but remains underrecognized in the community, despite the emergence of effective treatments.^{1,2} Different pathophysiologic phenotypes of HFpEF exist that are associated with greater risk, including patients with atrial myopathy,³⁻⁵ pulmonary vascular disease (PVD),^{6,7} and right ventricular (RV) failure.^{8,9} Although these subgroups of patients may be considered as phenotypes because of differing pathophysiologic mechanisms (and potentially, treatment targets), it also has been proposed that they may simply represent different stages of disease progression.¹⁰ If true, this would increase the priority for diagnosing HFpEF early in its course, to implement therapy earlier and thus slow or even prevent disease progression.

Although some studies have evaluated longitudinal changes in cardiac structure and function over time in HFpEF,^{8,11} little is known about how cardiac structure, function, and risk for adverse outcomes differ by duration of HF

diagnosis. In the REDUCE LAP-HF II (A Study to Evaluate the Corvia Medical, Inc IASD System II to Reduce Elevated Left Atrial Pressure in Patients with Heart Failure; [NCT03088033](#)) trial, implantation of the Corvia Atrial Shunt (IASD System II, Corvia Medical) in patients with HFpEF/HFmrEF had no significant effect on the primary composite outcome of cardiovascular death or ischemic stroke, HF events, and health status.¹² However, participants without latent PVD, with less right atrial (RA) remodeling, female sex, and without cardiac rhythm management devices showed signal of potential benefit, findings that could be related to shorter duration of HF before enrollment.¹³ Here, we tested the hypotheses that longer duration of HFpEF/HFmrEF would be associated with more severe cardiac remodeling and dysfunction, and greater risk for worsening HF. We further tested whether those with shorter duration of HF might have more favorable responses to atrial shunt, using baseline and post randomization data from the REDUCE LAP-HF II trial cohort.

METHODS

STUDY DESIGN. The REDUCE LAP-HF II study protocol, description, and primary findings have been previously reported.^{12,14} Eligible patients were

From the ^aDepartment of Cardiovascular Disease, Mayo Clinic, Rochester, Minnesota, USA; ^bDivision of Cardiology, Northwestern University Feinberg School of Medicine, Chicago, Illinois, USA; ^cCorvia Medical, Tewksbury, Massachusetts, USA; ^dArizona Heart Rhythm Center, Phoenix, USA; ^eDepartment of Internal Medicine, BG Klinikum Unfallkrankenhaus, Berlin, Germany; ^fDepartment of Cardiology, Campus Kerckhoff of the Justus-Liebig-University Giessen, Kerckhoff-Clinic, Bad Nauheim, Germany; ^gDivision of Cardiology, Medical University of South Carolina, Charleston, South Carolina, USA; ^hRalph H. Johnson Veterans Affairs Health System, Charleston, South Carolina, USA; ⁱScripps Clinic, La Jolla, California, USA; ^jDivision of Cardiovascular Medicine, The Ohio State University, Columbus, Ohio, USA; ^kCardiovascular Department, John Hunter Hospital, University of Newcastle, New Lambton Heights, Australia; ^lDepartment of Cardiovascular Medicine, Massachusetts General Hospital, Harvard Medical School, Boston, Massachusetts, USA; ^mDepartment of Cardiology, Alfred Hospital, Melbourne, Australia; ⁿSchool of Public Health and Preventive Medicine, Monash University, Melbourne, Australia; ^oMonash-Alfred-Baker Centre for Cardiovascular Research, Melbourne, Australia; ^pBaker Heart and Diabetes Institute, Melbourne, Australia; ^qDepartment of Cardiology, Centre for Congenital Heart Disease, University of Groningen, University Medical Centre Groningen, Groningen, the Netherlands; ^rUniversity of Michigan and VA Ann Arbor Health System, Ann Arbor, Michigan, USA; ^sDepartment of Cardiovascular Diseases, University of Zagreb School of Medicine, University Hospital Centre Zagreb, Zagreb, Croatia; ^tDepartment of Heart and Lung Transplantation, St Vincent's Hospital, Sydney, Australia; ^uDepartment of Cardiovascular Medicine, Kaufman Center for Heart Failure Treatment and Recovery, Heart, Vascular, and Thoracic Institute, Cleveland Clinic, Cleveland, Ohio, USA; ^vDivision of Cardiovascular Medicine, Vanderbilt University Medical Center, Nashville, Tennessee, USA; ^wDepartment of Internal Medicine, Division of Cardiology, Medical University of Graz, Graz, Austria; ^xThe Prince Charles Hospital, School of Medicine, University of Queensland, Brisbane, Queensland, Australia; ^yDepartment of Cardiology, Kerckhoff-Klinik, Bad Nauheim, Germany; ^zGerman Center for Cardiovascular Research, Partner Site RheinMain, Frankfurt AM Main, Germany; ^{aa}Baim Clinical Research Institute, Boston, Massachusetts, USA; ^{bb}Brigham and Women's Hospital, Boston, Massachusetts, USA; ^{cc}University Medical Center Groningen, University of Groningen, Groningen, the Netherlands; ^{dd}Department of Internal Medicine, Columbia University and Cardiovascular Research Foundation, New York City, New York, USA; and the ^{ee}Division of Cardiovascular Medicine, University of Pennsylvania, Philadelphia, Pennsylvania, USA.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

>40 years of age and had symptomatic HF, an ejection fraction $\geq 40\%$, and qualifying exercise right-sided heart catheterization demonstrating pulmonary capillary wedge pressure (PCWP) during exercise of ≥ 25 mm Hg that was ≥ 5 mm Hg above RA pressure. A comprehensive list of inclusion and exclusion criteria is available in the study design paper.¹⁴ Written informed consent was obtained from all trial participants before enrollment. The study was approved by the local ethics committees or Institutional Review Boards at each enrolling site.

Duration of HFpEF/HFmrEF diagnosis data were collected prospectively in the case report forms at the time of enrollment. Patients who did not have duration of HF data were excluded from this analysis. Participants were categorized according to the duration of diagnosis, divided into 3 groups: diagnosis <1 year, diagnosis 1 to 3 years, and diagnosis >3 years. Duration of diagnosis was entered by site staff into the case report form based on their medical history-taking and chart review. Comparison of baseline characteristics according to the duration of HFpEF/HFmrEF diagnosis was performed, along with risk for HF outcomes and evaluation of the response to atrial shunt therapy as compared to sham.

STUDY PROTOCOL. Participants underwent comprehensive echocardiography and invasive hemodynamic exercise testing in the supine position before randomization to confirm eligibility. Pressures were measured in a blinded fashion at end-expiration at rest and during exercise at a central core laboratory, taking the average of ≥ 3 beats. As previously described, eligible participants were then randomly assigned (1:1) to receive the Corvia Atrial Shunt or a sham procedure.^{12,14} Echocardiograms and invasive hemodynamic pressure tracings were interpreted by an independent core laboratory blinded to all other data. Echocardiographic measurements were performed according to current guidelines in a separate core lab.^{15,16}

Left ventricular (LV) and left atrial (LA) volumes were measured in 2- and 4-chamber views using the Simpson biplane method. RA volumes were obtained in the apical 4-chamber view. Tissue Doppler of the mitral annulus was performed to measure early diastolic (e') and systolic (s') tissue velocities. RV systolic function was assessed by tricuspid annular plane systolic excursion (TAPSE) and tricuspid lateral annular plane systolic excursion (RV s'). Speckle tracking was used for strain measurements of all 4 cardiac chambers in the apical 4-chamber view using TomTec Cardiac Performance Analysis software at the Northwestern

University Echocardiography Core Laboratory (Chicago, Illinois, USA). LV global longitudinal strain (GLS) was measured using 2-dimensional speckle tracking according to current standards.^{15,17} RV GLS and RV free wall longitudinal strain were obtained by the average values of peak longitudinal systolic strain from the free wall and septal wall of the RV.^{8,18} The myocardial reservoir function of the LA was evaluated by the LA strain peak during LA relaxation.^{18,19} With the use of 2-dimensional echocardiographic speckle tracking, RA endocardial contours were traced, and RA reservoir strain was subsequently measured.¹⁸ LA and RA myopathy were defined as LA volume index of >34 mL/m² and/or LA reservoir strain of $\leq 24\%$, RA volume index of >39 mL/m² in men or >33 mL/m² in women, and/or RA reservoir strain $\leq 19.8\%$, respectively.^{5,15,20}

Clinicians conducted follow-up visits masked to treatment allocation at 6, 12, and 24 months, at which time assessments were performed for worsening HF events, adverse events, health status, and the Kansas City Cardiomyopathy Questionnaire-Overall Summary Score (KCCQ-OSS). Patients were followed for up to 5 years.

TRIAL OUTCOMES. The primary endpoint was the hierarchical composite endpoint of cardiovascular death or nonfatal ischemic stroke (over the maximal duration of follow-up), first and recurrent HF hospitalization or urgent intensification of oral diuresis (over the maximal duration of follow-up), and change in KCCQ-OSS between baseline and 12 months, analyzed when the last randomized patient reached 12 months of follow-up. The secondary efficacy endpoints included the individual components of the hierarchical composite events. All events were adjudicated by a blinded clinical events committee.

SENSITIVITY ANALYSIS. As a sensitivity analysis, we evaluated an independent Mayo Clinic HFpEF cohort (all ejection fractions $\geq 50\%$) participating in a separate prospective study²¹ in which patients were categorized according to duration of HF diagnosis (<1 year, 1 to 3 years, and >3 years) (n = 50 in each group). Baseline clinical characteristics, echocardiographic parameters, and invasive hemodynamic measurements at rest and during exercise were compared across duration groups using the same general analytic approach.

STATISTICAL ANALYSIS. Patient characteristics are presented as numbers (percentage) for categorical variables and as median (Q1-Q3) for continuous variables. For continuous and categorical variables,

differences between the 3 groups (diagnosis <1 year, 1-3 years, and >3 years) were assessed using linear regression to examine the trend across the 3 categories. The Jonkheere-Terpstra test was also performed to assess for ordered trends. Statistical significance was defined as a 2-sided value of $P < 0.05$. In calculating the win ratio, all patients are compared pairwise with each other on the values of the 3 components in a hierarchical manner.²² Additional analyses included individual examination of components of the primary endpoint (incident cardiovascular death or nonfatal ischemic stroke [Cox proportional hazards analysis], recurrent events analysis for HF events [negative binomial regression adjusted for number of HF hospitalizations in the year before randomization], and change in KCCQ-OSS from baseline to 12 months using linear regression with adjustment for the baseline KCCQ-OSS value) calculated for each randomized treatment group within the duration of diagnosis groups. The incidence of the secondary endpoints was presented for each treatment group, with treatment group comparisons performed on the incidence rate using logistic regression within the subgroup. The incidence of HF events was compared between groups from Kaplan-Meier estimates.

We analyzed the interaction effect between HF duration and cardiovascular death or nonfatal ischemic stroke, HF events, or change in KCCQ score in 2 ways: first, as a continuous variable (per 1-year increase in the log duration of diagnosis), and second, as an ordinal variable (diagnosis <1 year, 1-3 years, and >3 years). A test for equality of the win ratio across the 3 HF duration categories (modeled as an ordinal variable to test for trend) was performed using meta-regression. We used linear regression analysis to assess the effect of shunt therapy on 12-month and 24-month changes in echocardiographic parameters, adjusted for the baseline value. Mean (95% CI) changes in 12-month and 24-month echocardiographic parameters were reported for each group. We also calculated P values for the interaction of treatment $\times$ HF duration categories on changes in 12- and 24-month echocardiographic parameters. Analyses were performed using Stata version 17.0 (StataCorp).

RESULTS

Of the 626 participants randomized, 522 had data on the duration of diagnosis captured and were included in the primary analysis. Compared with those without a known duration of HF diagnosis, individuals with a known duration of diagnosis were

more likely to be female and had a lower prevalence of diabetes and atrial flutter, as well as lower NYHA functional class and hemoglobin levels (Supplemental Table 1). Participants displayed clinical characteristics typical of HFpEF/HFmrEF, including advanced age (72 years [Q1-Q3: 66-77 years]) and female predominance (63.2%). Most patients had HFpEF (82.5%), and the minority had HFmrEF (17.5%). In the patients with a known duration of HF diagnosis, 211 (40.4%) were diagnosed within 1 year of enrollment, 152 (29.1%) were diagnosed within 1 to 3 years, and 159 (30.5%) were diagnosed >3 years before enrollment.

Baseline characteristics according to the duration of diagnosis are shown in Table 1. Participants with longer duration of diagnosis were more likely to be men and had higher prevalence of ischemic heart disease, myocardial infarction, and HFmrEF, greater use of loop diuretic agents and beta-blockers, less use of renin-angiotensin system blockers, and higher N-terminal pro-B-type natriuretic peptide levels compared with those with shorter duration of diagnosis. Patient-reported quality of life was severely impaired across all durations of HF (Table 1).

CARDIAC STRUCTURE, FUNCTION, AND HEMODYNAMICS.

As compared with those with a shorter duration of diagnosis, patients with HFpEF/HFmrEF and a longer duration of diagnosis had more profound cardiac remodeling and dysfunction (Table 2, Figure 1). Patients with biatrial myopathy also had a higher prevalence of atrial fibrillation/flutter, cardiac rhythm management devices, and higher natriuretic peptide levels (Supplemental Table 2). Specifically, patients with longer duration of diagnosis had higher LV mass index and LV end-diastolic volume, more severely impaired LV GLS and diastolic function, greater LA remodeling, and more severely impaired LA reservoir strain. Similarly, RV function was more impaired in those with longer duration of treatment, with lower RV free wall strain, TAPSE, and RV s' velocity. This was coupled with more advanced RA remodeling and more impaired RA reservoir strain in those with longer duration of HFpEF/HFmrEF diagnosis compared with shorter duration.

As compared with those with shorter duration of diagnosis, patients with longer duration had higher PCWP at rest along with higher mean pulmonary artery (PA) pressure, total pulmonary resistance, and pulmonary elastance, with more impaired RV-PA coupling (lower TAPSE/PA systolic pressure). Elevation in PCWP restricted to exercise (normal at rest) was more common in those with shorter duration of diagnosis (35.1% vs 29.3% vs 24.2% for <1 year, 1-3

TABLE 1 Baseline Characteristics by Duration of HF Diagnosis Duration

	Diagnosis <1 y (n = 211)	Diagnosis 1-3 y (n = 152)	Diagnosis >3 y (n = 159)	P Value
Age, y	72 (66-77)	72 (65-77)	73 (66-78)	0.70
Female	148 (70.1)	96 (63.2)	86 (54.1)	0.002
Body mass index, kg/m ²	31.4 (27.2-36.9)	32.1 (27.9-36.9)	32.8 (28.0-37.2)	0.27
Comorbidities				
Hypertension	180 (86.1)	135 (88.8)	142 (89.3)	0.33
Diabetes	60 (28.4)	62 (40.8)	61 (38.4)	0.030
COPD	38 (18.0)	30 (19.7)	37 (23.3)	0.22
Ischemic heart disease	21 (10.0)	26 (17.3)	34 (21.4)	0.002
Myocardial infarction	20 (9.6)	20 (13.2)	33 (20.8)	0.003
Prior PCI or CABG	48 (23.0)	42 (27.8)	52 (32.7)	0.038
Prior valve surgery/intervention	6 (6.6)	13 (8.8)	13 (16.2)	0.053
Cardiac rhythm management (pacemaker/ICD/CRT)	32 (15.2)	27 (17.8)	34 (21.7)	0.11
Atrial fibrillation or flutter	104 (49.3)	83 (54.6)	96 (60.4)	0.034
HF status				
NYHA functional class III	153 (73.2)	112 (75.7)	129 (81.6)	0.066
Number of hospitalizations in the past 12 mo for HF	1.0 (0.5-1.0)	1.0 (0.0-1.0)	1.0 (0.0-1.5)	0.86
KCCQ-OSS at baseline	46.6 (28.6-65.6)	43.5 (27.5-62.8)	46.6 (32.0-63.5)	0.97
HF subtype				0.005
HFmrEF (EF 41%-49%)	22 (12.3)	23 (16.9)	35 (24.5)	
HFpEF (EF ≥50%)	157 (87.7)	113 (83.1)	108 (75.5)	
No latent PVD or cardiac rhythm device (n = 486)	115 (58.1)	85 (59.0)	71 (49.3)	0.14
Medications				
Loop diuretics	153 (72.5)	136 (89.5)	139 (87.4)	0.0001
Loop diuretic dose, mg (furosemide equivalents)	40.0 (20.0-40.0)	40.0 (20.0-80.0)	40.0 (20.0-80.0)	0.049
Beta-blockers	134 (63.5)	104 (68.4)	125 (78.6)	0.002
ACEI/ARB/ARNI	149 (70.6)	87 (57.2)	89 (56.0)	0.002
MRA	111 (52.6)	81 (53.3)	85 (53.5)	0.87
SGLT2 inhibitors	6 (2.8)	2 (1.3)	8 (5.0)	0.31
Laboratories				
Hemoglobin, g/dL	12.8 (11.6-13.9)	12.7 (11.2-13.7)	12.8 (11.6-13.9)	0.97
eGFR	58.4 (44.0-69.0)	56.0 (39.0-69.0)	55.0 (43.6-63.0)	0.11
BNP, pg/mL	80 (37-147)	110 (50-220)	148 (72-259)	0.0003
NT-proBNP, pg/mL	267 (137-605)	548 (251-1,085)	536 (252-1,214)	<0.0001

Values are median (Q1-Q3) or n (%), unless otherwise indicated.

ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor neprilysin inhibitor; BNP = b-type natriuretic peptide; CABG = coronary artery bypass graft; COPD = chronic obstructive pulmonary disease; CRT = cardiac resynchronization therapy; eGFR = estimated glomerular filtration rate; EF = ejection fraction; HF = heart failure; HFmrEF = heart failure with mildly reduced ejection fraction; HFpEF = heart failure with preserved ejection fraction; ICD = implantable cardioverter-defibrillator; KCCQ-OSS = Kansas City Cardiomyopathy Questionnaire-Overall Summary Score; MRA = mineralocorticoid receptor antagonist; NT-proBNP = N-terminal pro-B-type natriuretic peptide; PCI = percutaneous coronary intervention; PVD = pulmonary vascular disease; SGLT2 = sodium-glucose cotransporter 2.

years, and >3 years, respectively; $P = 0.025$). There were no differences in RA pressure or cardiac output at rest. There was higher PCWP during exercise in patients with longer duration of HF diagnosis, but there were no other significant differences in any hemodynamic measures during exercise (Table 2).

DURATION OF HF DIAGNOSIS AND RISK FOR WORSENING HF. During a median follow-up duration of 1,541 days (Q1-Q3: 1,179-1,828 days), the HF event rate increased with duration of diagnosis in the sham group, from 15.5 per 100 patient-years in those with diagnosis <1 year, to 17.7 per 100 patient-years with diagnosis 1 to 3 years, to 19.0 per 100 patient-years in those with diagnosis for

>3 years (Table 3). Kaplan-Meier analysis for HF events indicated that the risk of HF events were higher in patients with a longer duration of diagnosis than those with a shorter duration of diagnosis (HR: 1.70 [95% CI: 1.17-2.48]; $P = 0.006$, for duration >3 years vs <1 year [Reference]) (Figure 2). Similar findings were observed within previously identified responder and nonresponder subgroups, with no significant treatment-by-duration interactions (Supplemental Table 3).

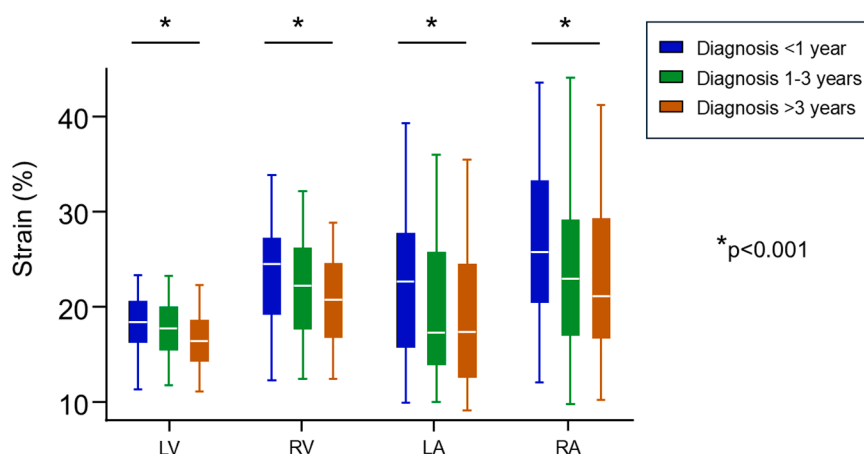
EFFECTS OF THE DURATION OF DIAGNOSIS ON RESPONSE TO SHUNT THERAPY. There was no statistically significant heterogeneity in the response to atrial shunt treatment based on the duration of

TABLE 2 Cardiac Structure, Function, and Hemodynamics at Baseline by HF Diagnosis Duration

	Diagnosis <1 y (n = 211)	Diagnosis 1-3 y (n = 152)	Diagnosis >3 y (n = 159)	P Value
LV structure/function				
LV ejection fraction, %	55.2 (50.6-58.0)	54.4 (50.3-57.0)	52.2 (47.7-56.5)	0.0008
LV mass index, g/m ²	77.1 (63.9-91.6)	79.7 (66.2-100.2)	86.5 (71.4-106.3)	0.0002
LV end-diastolic volume index, mL/m ²	52.1 (45.0-65.0)	51.9 (43.0-61.1)	56.8 (49.0-65.0)	0.044
E/A ratio	1.0 (0.8-1.4)	1.1 (0.8-1.7)	1.2 (0.8-1.7)	0.036
Average E/e' ratio	11.7 (9.7-14.7)	13.2 (9.3-19.1)	13.6 (9.6-17.0)	0.022
LV GLS, %	18.4 (16.3-20.5)	17.7 (15.4-20.0)	16.4 (14.3-18.6)	<0.0001
LA structure/function				
LA maximal volume index, mL/m ²	29.5 (24.0-36.2)	29.9 (24.2-37.7)	34.0 (26.4-40.9)	0.003
LA minimal volume index, mL/m ²	18.1 (14.3-24.6)	19.5 (14.5-26.8)	22.9 (15.7-29.8)	0.001
LA reservoir strain, %	22.6 (15.8-27.7)	17.3 (13.8-25.7)	17.4 (12.6-24.4)	0.0003
RV structure/function				
RV end-diastolic volume, mL	40.0 (30.5-51.0)	45.0 (34.0-58.0)	47.0 (36.6-62.0)	0.002
TAPSE, cm	2.1 (1.8-2.3)	2.0 (1.7-2.3)	2.0 (1.7-2.2)	0.006
Estimated PA systolic pressure, mm Hg	30.2 (25.5-35.0)	33.2 (26.2-40.5)	31.3 (26.0-39.0)	0.10
TAPSE/ePASP ratio, mm/mm Hg	0.7 (0.6-0.9)	0.6 (0.4-0.8)	0.6 (0.5-0.7)	0.001
RV s' velocity, cm/s	12.0 (10.0-14.0)	11.0 (9.0-14.0)	11.0 (9.0-13.0)	0.0005
RV free wall strain, %	24.4 (19.2-27.2)	22.2 (17.6-26.1)	20.8 (16.7-24.6)	<0.0001
RA structure/function				
RA volume index, mL/m ²	22.6 (17.7-28.7)	24.5 (18.8-32.1)	27.2 (21.0-36.5)	0.0001
RA reservoir strain, %	25.7 (20.5-33.2)	22.9 (17.0-28.8)	21.1 (16.7-29.3)	0.0001
Valvular disease				
Moderate or greater MR	10/173 (5.8)	15/137 (11.0)	13/137 (9.5)	0.15
Moderate or greater TR	17/175 (9.7)	27/136 (19.9)	15/136 (11.0)	0.07
Atrial myopathy				
No atrial myopathy	55 (33.7)	22 (18.3)	23 (18.1)	<0.0001
Isolated RA myopathy	3 (1.8)	3 (2.5)	6 (4.7)	
Isolated LA myopathy	59 (36.2)	46 (38.3)	36 (28.3)	
Biatrial myopathy	46 (28.2)	49 (40.8)	62 (48.8)	
Resting hemodynamics				
Heart rate, beats/min	68 (61-78)	70 (62-80)	70 (64-79)	0.18
Aortic systolic pressure, mm Hg	145 (130-159)	142 (127-157)	143 (128-156)	0.33
RA pressure, mm Hg	9 (6-11)	9 (6-12)	9 (7-12)	0.08
Mean PA pressure, mm Hg	24 (21-29)	26 (22-33)	27 (23-33)	0.005
PCWP, mm Hg	16 (13-22)	17 (14-22)	18 (15-23)	0.008
Resting PCWP ≥15	135 (64.9)	106 (70.7)	119 (75.8)	0.024
Cardiac output, L/min	5.2 (4.4-6.2)	5.1 (4.4-6.2)	5.1 (4.3-6.2)	0.32
Stroke volume, mL	75 (64-91)	75 (63-88)	73 (63-86)	0.19
PVR, WU	1.4 (1.1-2.0)	1.6 (1.1-2.1)	1.6 (1.0-2.2)	0.30
Total pulmonary resistance, WU	4.6 (3.7-6.0)	5.1 (3.8-6.5)	5.2 (4.1-6.7)	0.004
PA pulse pressure/stroke volume, mm Hg/mL	0.9 (0.7-1.2)	0.8 (0.7-1.1)	0.9 (0.7-1.1)	0.53
RV effective arterial elastance, mm Hg/mL	0.5 (0.4-0.6)	0.6 (0.4-0.8)	0.5 (0.4-0.7)	0.02
TAPSE/PA systolic pressure ratio, mm/mm Hg	0.6 (0.4-0.7)	0.5 (0.4-0.6)	0.5 (0.4-0.6)	0.002
Peak exercise hemodynamics				
Heart rate, beats/min	102 (86-114)	103 (87-114)	98 (86-110)	0.23
Aortic systolic pressure, mm Hg	163 (140-184)	157 (142-180)	159 (141-177)	0.20
RA pressure, mm Hg	17 (14-22)	17 (14-22)	18 (15-23)	0.07
Mean PA pressure, mm Hg	45 (40-52)	45 (40-51)	47 (40-52)	0.34
PCWP, mmHg	33 (28-38)	34 (29-41)	35 (31-42)	0.012
Cardiac output, L/min	8.2 (6.6-10.5)	8.0 (6.3-9.9)	7.8 (6.2-9.7)	0.16
Stroke volume, mL	83 (67-105)	79 (63-98)	82 (64-100)	0.89
PVR, WU	1.3 (0.9-2.0)	1.3 (0.8-1.9)	1.4 (0.8-2.0)	0.63
PVR at peak exercise <1.75 WU	138 (69.7)	99 (70.2)	90 (64.3)	0.34
Total pulmonary resistance, WU	5.5 (4.3-7.1)	5.6 (4.6-7.2)	6.0 (4.4-7.8)	0.14
PA pulse pressure/stroke volume, mm Hg/mL	1.4 (1.0-1.8)	1.4 (1.0-1.9)	1.3 (1.0-1.8)	0.78
RV effective arterial elastance, mm Hg/mL	1.3 (1.0-1.7)	1.3 (1.0-1.6)	1.3 (1.0-1.6)	0.86
Peak workload, Watts	40 (20-60)	40 (20-60)	40 (20-60)	0.33

Values are median (Q1-Q3), n/N (%), or n (%), unless otherwise indicated.

E/A = early to late diastolic transmitral flow velocity; E/e' = early diastolic mitral annular tissue velocity; ePASP = estimated pulmonary artery systolic pressure by echocardiography; GLS = global longitudinal strain; LA = left atrial; LV = left ventricular; MR = mitral regurgitation; PA = pulmonary artery; PCWP = pulmonary capillary wedge pressure; PVR = pulmonary vascular resistance; RA = right atrial; RV = right ventricular; TAPSE = tricuspid annular plane systolic excursion; TR = tricuspid regurgitation.

FIGURE 1 4-Chamber Cardiac Mechanics by Duration of HF Diagnosis

Left ventricular (LV) global longitudinal strain, right ventricular (RV) free wall strain, left atrial (LA) reservoir strain, and right atrial (RA) reservoir strain were lower in patients with a longer duration of diagnosis. HF = heart failure.

diagnosis (Table 3, Central Illustration). In patients with a diagnosis <1 year, atrial shunt treatment was associated with nonsignificant decreases in the risk of worsening HF (HR: 0.74 [95% CI: 0.34-1.63]; $P = 0.46$) and nonsignificant improvements in KCCQ-OSS (+3.9 compared with sham [95% CI: -1.3 to +9.2]; $P = 0.14$). Conversely, in patients with a diagnosis >3 years, atrial shunt treatment was

associated with an increased risk for worsening HF events (HR: 1.72 [95% CI: 1.26-2.35]; $P = 0.001$) and nonsignificant worsening in KCCQ-OSS (-2.1 [95% CI: -8.7 to +4.5]; $P = 0.62$) compared with sham.

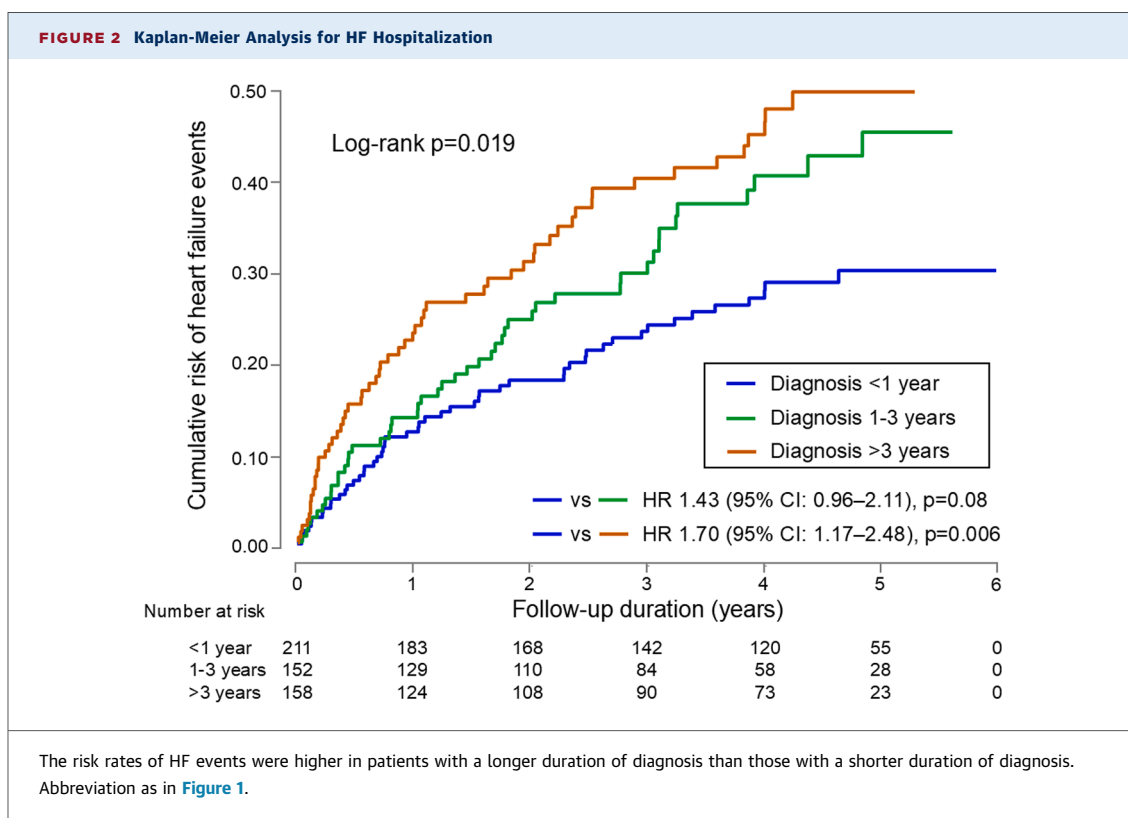
SENSITIVITY ANALYSIS IN THE MAYO CLINIC COHORT. In the Mayo Clinic cohort ($n = 150$), longer HFpEF duration was similarly associated with greater

TABLE 3 Efficacy Outcomes Stratified by HF Duration

Statistic		Atrial Shunt	Sham Control	Win Ratio, HR, IRR, or Estimated Treatment Difference (95% CI)	P Value ^a
Diagnosis <1 y					
Composite primary endpoint	Win ratio	—	—	1.18 (0.85-1.64)	0.32
CV death/nonfatal stroke	Rate per 100 patient-years	1.91	1.00	1.87 (0.61-5.73)	0.27
HF event ^b	Rate per 100 patient-years	10.2	15.5	0.74 (0.34-1.63)	0.46
ΔKCCQ-OSS at 12 mo	Mean (95% CI)	15.4 (11.2-19.7) ($n = 95$)	11.5 (7.7-15.3) ($n = 110$)	+3.9 (-1.3 to +9.2)	0.14
Diagnosis 1-3 y					
Composite primary endpoint	Win ratio	—	—	1.04 (0.71-1.53)	0.83
CV death/nonfatal stroke	Rate per 100 patient-years	4.09	3.48	1.14 (0.50-2.56)	0.76
HF event ^b	Rate per 100 patient-years	15.7	17.7	0.77 (0.40-1.48)	0.44
ΔKCCQ-OSS at 12 mo	Mean (95% CI)	9.2 (4.1-14.3) ($n = 77$)	7.6 (2.0-13.2) ($n = 63$)	+1.3 (-5.1 to +7.8)	0.69
Diagnosis >3 y					
Composite primary endpoint	Win ratio	—	—	0.83 (0.57-1.20)	0.31
CV death/nonfatal stroke	Rate per 100 patient-years	3.70	1.63	2.20 (0.81-5.95)	0.53
HF event ^b	Rate per 100 patient-years	22.2	19.0	1.72 (1.26-2.35)	0.001
ΔKCCQ-OSS at 12 mo	Mean (95% CI)	8.3 (2.7-13.8) ($n = 67$)	10.8 (6.0-15.5) ($n = 79$)	-2.1 (-8.7 to +4.5)	0.62

^aP value for CV death/stroke calculated using Cox regression and HR reported; P value for HF event calculated using negative binomial regression (for total [first and recurrent] HF events) adjusted for number of HF hospitalizations in 12 months before randomization (with IRR reported); P value for change in KCCQ-OSS calculated using linear regression adjusting for baseline KCCQ-OSS value. ^bHF event includes HF hospitalizations and urgent outpatient visit for worsening HF requiring either intravenous diuretic agents or oral diuretic intensification.

CV = cardiovascular; IRR = incidence rate ratio; other abbreviations as in Table 1.



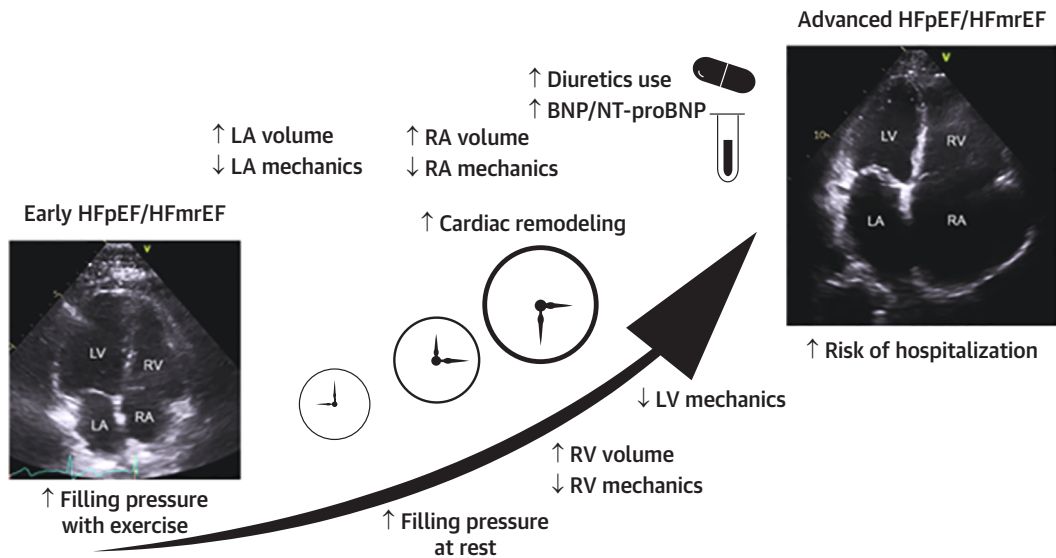
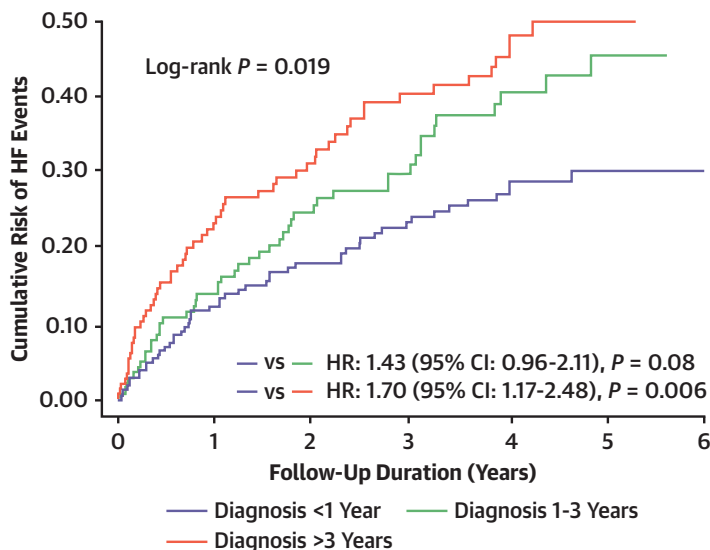
disease severity across multiple domains, including worse LV remodeling and lower LV GLS, greater LA remodeling and dysfunction, worse RV systolic function, higher resting LV filling pressures and PA pressures, lower resting cardiac output, and impaired exercise hemodynamic reserve, with higher exercise pulmonary vascular resistance ([Table 4](#), [Supplemental Table 4](#)), confirming and extending on the primary results.

DISCUSSION

In the REDUCE LAP-HF II trial, patients with a longer duration of HF had more severe cardiac remodeling and dysfunction across multiple metrics, were more likely to have hemodynamic congestion that was apparent even at rest, and displayed greater risk of HF events, compared with those with shorter duration of diagnosis. There was no statistically significant heterogeneity in response to atrial shunt treatment based on the duration of diagnosis. More adverse cardiac remodeling, dysfunction, and greater hemodynamic severity were also demonstrated in an independent validation cohort. By showing the progressive deterioration in cardiac structure and function with longer HF duration, these data emphasize

the importance of earlier diagnosis of HFpEF/HFmrEF in patients with dyspnea, enabling introduction of foundational medical treatments at an earlier stage of disease, before cardiac remodeling and dysfunction worsen to a greater extent, which was in turn, associated with a greater risk for subsequent worsening HF events ([Central Illustration](#)).

The most striking differences in cardiac structure and function based on HF duration related to biatrial dysfunction and remodeling, with longer diagnosis duration patients having more advanced biatrial myopathy, and greater biventricular dysfunction and remodeling, including more severely abnormal LV and RV mechanics by speckle tracking strain imaging. Prior studies have shown that impairments in atrial,^{5,23,24} LV,²⁵ and RV function^{8,9} are associated with greater risk for HF hospitalization or death. Patients with greater atrial myopathy or right HF have been considered to represent discrete pathophysiologic phenotypes because of these differences in pathophysiology and risk.^{26,27} It has been speculated that these pathophysiologic characteristics also may reflect greater duration of HF, leading to more advanced cardiac remodeling and dysfunction.¹⁰ The present data support the concept of greater disease duration leading to more advanced phenotypes,

CENTRAL ILLUSTRATION HF Duration, Cardiac Remodeling, and Hemodynamic Severity in HFpEF/HFmrEF**Clinical and Pathophysiologic Progression With Longer HF Duration****Risk of HF Events by HF Diagnosis Duration**

Doi S, et al. JACC Heart Fail. 2026;■(■):103167.

Summary of the key findings of the study relating heart failure (HF) duration to pathophysiology and risk for HF events. BNP = B-type natriuretic peptide; HFmrEF = heart failure with mildly reduced ejection fraction; HFpEF = heart failure with preserved ejection fraction; LA = left atrium; LV = left ventricle; NT-proBNP = N-terminal pro-B-type natriuretic peptide; RA = right atrium; RV = right ventricle.

TABLE 4 Cardiac Structure, Function, and Hemodynamics at Baseline by HFpEF Diagnosis Duration in Mayo Cohort

	Diagnosis <1 y (n = 50)	Diagnosis 1-3 y (n = 50)	Diagnosis >3 y (n = 50)	P Value
LV structure/function				
LV ejection fraction, %	63 (59-68)	61 (58-64)	61 (58-66)	0.074
LV mass index, g/m ²	73.7 (64.8-85.3)	76.0 (71.4-85.5)	83.0 (73.5-92.9)	0.008
LV end-diastolic volume index, mL/m ²	59.3 (47.1-78.6)	59.2 (46.6-73.3)	57.4 (47.9-67.0)	0.35
E/A ratio	0.9 (0.8-1.0)	0.9 (0.8-1.2)	1.0 (0.8-1.8)	0.089
Average E/e' ratio	8.8 (7.3-13.3)	9.5 (7.8-12.3)	10.6 (8.9-14.4)	0.011
LV GLS, %	18.7 (17.2-21.7)	17.5 (16.8-20.2)	16.9 (15.7-19.5)	0.006
LA structure/function				
LA maximal volume index, mL/m ²	21.5 (17.0-29.5)	26.8 (19.8-35.0)	30.6 (22.1-44.0)	0.002
LA minimal volume index, mL/m ²	11.3 (8.9-15.2)	12.5 (9.4-17.4)	15.7 (10.7-28.7)	0.003
LA reservoir strain, %	23.1 (19.5-25.8)	22.3 (18.9-25.4)	19.7 (13.4-22.6)	0.001
RV structure/function				
TAPSE, cm	2.4 (2.0-2.7)	2.2 (2.1-2.5)	1.9 (1.2-2.4)	0.003
RV s' velocity, cm/s	12.0 (9.0-15.0)	11.0 (10.0-12.0)	10.0 (7.0-12.0)	0.003
RV free wall strain, %	27.5 (25.2-31.8)	26.3 (24.1-30.4)	22.4 (21.3-24.2)	<0.0001
Resting hemodynamics				
Heart rate, beats/min	71 (64-80)	70 (62-75)	67 (65-75)	0.27
Aortic systolic pressure, mm Hg	149 (126-176)	142 (133-155)	138 (126-151)	0.063
RA pressure, mm Hg	8 (6-10)	8 (6-10)	9 (6-12)	0.13
Mean PA pressure, mm Hg	22 (19-26)	23 (20-25)	25 (20-30)	0.041
PCWP, mm Hg	13 (10-16)	13 (10-17)	15 (11-20)	0.017
Cardiac output, L/min	5.4 (4.7-6.5)	5.2 (4.7-6.2)	5.0 (4.4-5.5)	0.020
Stroke volume, mL	81 (63-91)	78 (67-85)	74 (66-87)	0.61
PVR, WU	1.5 (1.2-2.0)	1.6 (1.0-2.1)	1.9 (1.3-2.9)	0.066
Peak exercise hemodynamics				
Heart rate, beats/min	117 (104-124)	108 (100-126)	107 (94-125)	0.061
Aortic systolic pressure, mm Hg	197 (180-229)	189 (169-212)	190 (173-206)	0.10
RA pressure, mm Hg	17 (14-20)	17 (13-21)	19 (15-25)	0.11
Mean PA pressure, mm Hg	40 (36-50)	40 (33-46)	41 (35-51)	0.86
PCWP, mm Hg	31 (28-36)	30 (26-35)	30 (25-34)	0.093
Cardiac output, L/min	10.3 (8.7-12.5)	10.8 (8.7-13.3)	8.6 (6.8-11.4)	0.012
Stroke volume, mL	96 (78-108)	95 (78-122)	80 (66-103)	0.040
PVR, WU	0.9 (0.4-1.4)	1.2 (0.9-1.7)	2.0 (1.2-5.5)	0.001
Peak workload, Watts	80 (40-100)	80 (60-120)	70 (33-100)	0.91

Values are median (Q1-Q3) or n (%), unless otherwise indicated.
Abbreviations as in Table 2.

emphasizing the importance for earlier diagnosis and treatment. Patients with shorter duration of diagnosis were also more likely to have elevated PCWP that was only apparent during exercise, being absent at rest, further supporting the idea that hemodynamic abnormalities also worsen over time in HFpEF/HFmrEF. In this regard, it has also recently been shown that approximately 40% of patients with HFpEF and normal resting PCWP progress to develop resting congestion over time,²⁸ further supporting a continuum of early to later stages of disease progression.

HF is characterized by an inability of the heart to provide sufficient cardiac output at a rate commensurate with the body's needs without requiring high filling pressures, at rest or with exercise.²⁹ Over time,

chronic, sustained elevation in filling pressures is thought to cause cardiac remodeling and dysfunction.^{23,24,30} The present findings support this concept, whereby HF and its hemodynamic sequelae beget worsening HF over time.

Few studies have evaluated longitudinal changes in cardiac structure and function with repeated measurements in HFpEF. Obokata et al⁸ showed that worsening RV dysfunction and RV and RA dilatation commonly develop over a median follow-up of 4 years, with changes greatly exceeding differences in the LV. Atrial fibrillation was a strong predictor of incident RV dysfunction, although it remains unclear if the rhythm disturbance is causal, or a marker of more advanced atrial myopathy and HFpEF.

Sorimachi et al¹¹ also showed that over time there is progressive worsening in atrial and ventricular mechanics, assessed by speckle tracking strain, in patients with HFpEF. In a cross-sectional study, Omote et al⁵ showed that patients with greater LA and RA myopathy had multiple other features typical of more advanced HFpEF, including more severe RV dysfunction, PVD, tricuspid regurgitation, and a higher burden of atrial fibrillation, all findings similar to the group with longest duration of HF in the present study. The present findings in tandem with these earlier studies underline the clinical importance of advanced HFpEF and the importance for initiating treatments earlier before remodeling and dysfunction progresses to less reversible stages, which is even more important now given the emergence of effective treatments.

In REDUCE LAP-HF II, atrial shunt treatment had no effect on the hierarchical composite outcome in patients with HFpEF/HFmrEF.¹² Here, we observed no heterogeneity in the efficacy of shunt treatment in patients with shorter diagnosis duration, although there was signal suggesting beneficial trends in patients with shorter diagnosis duration, and signal of harm in those with longer duration. In REDUCE LAP-HF II, there was evidence for benefit in patients with no latent PVD, and a significant treatment interaction for PA systolic pressure during exercise in prespecified subgroup analyses.^{13,31,32} In healthy individuals, there is a reduction in pulmonary vascular resistance (PVR) during exercise, as flow-mediated dilation and vascular recruitment allows for increases in pulmonary blood flow without untoward increases in PA pressure. We previously showed that patients with HFpEF/HFmrEF that lack this ability (ie, high exercise PVR, termed latent PVD) respond poorly to atrial shunt therapy, whereas those with no latent PVD showed signal of benefit. Because PVD progresses with a longer duration of HF, and is associated with atrial myopathy, we speculated that patients with a shorter duration of HF might be more likely to display characteristics of the responder group, but there were no statistically significant differences in cardiac rhythm management device history or exercise PVR by duration of HF.

Right heart dilation is likely a necessary adaptation to shunt therapy, indicating that increases in pulmonary blood flow are present—a potential marker of successful shunt performance. We observed that RV dilation and RA enlargement without change of TAPSE in shunt treatment did not vary by duration of HF.

However, there was a signal for greater left heart plasticity and unloading in those with shorter duration of HF, as shunt therapy reduced LA minimal volume compared with sham at 12 months, and there was also a reduction in LV end-diastolic volume with shunt therapy compared with sham at 12 and 24 months in those with shorter duration of HF diagnosis, although formal tests for treatment-by-duration interaction were not significant.

STUDY LIMITATIONS. This is a post hoc analysis of the trial that was not designed specifically to evaluate the effects of shunt therapy according to duration of HF diagnosis. Although data on duration of HF diagnosis were systematically collected, there is likely inaccurate recall by patients during history-taking and based on chart review for the exact date of HF diagnosis. Further, symptoms develop indolently over time and may predate a clinical diagnosis for an extended period, making it difficult to identify a specific date. However, this limitation applies to all groups, so there is little bias related to this limitation.

CONCLUSIONS

Longer duration of HF diagnosis is associated with more advanced cardiac remodeling and dysfunction, greater hemodynamic disease severity, and a higher risk of worsening HF in patients with HFpEF/HFmrEF, but HF diagnosis duration did not modify the response to atrial shunt therapy. The greater remodeling, dysfunction, and risk for worsening HF observed in those with longer duration of HF raise the priority for earlier identification and treatment of HFpEF/HFmrEF.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

The REDUCE LAP-HF II trial was funded by Corvia. Dr Mohan has received consultancy fees from Alleviant Medical, AstraZeneca, Bayer, Boston Scientific, BMS, BridgeBio, Corvia, CVRx, Pfizer, and Viz.ai. Dr Sverdllov is supported by the National Heart Foundation of Australia Future Leader Fellowship (106025); has received research support from the Commonwealth Department of Health and Aged Care (Australia) Medical Research Future Fund, NSW Ministry of Health Cardiovascular Research Capacity Program, Cancer Institute NSW (Australia), as well as research grant funding from AstraZeneca, Biotronik, GlaxoSmithKline, Novartis, and Race Oncology; and has received speaker/consultancy fees from AstraZeneca, Boehringer Ingelheim, Cipla, Janssen, and Novartis. Dr Shah was supported by research grants from the National Institutes of Health (NIH) (U54 HL160273, R01 HL140731, R01 HL149423, X01 HL169712), American Heart Association 24SFRNPCN1291224), AstraZeneca, Boston Scientific, Corvia, Pfizer, and Tempus; and has received consulting fees

from 35Pharma, Abbott, Alleviant, AstraZeneca, Amgen, Aria CV, Axon Therapies, BaroPace, Bayer, Boehringer Ingelheim, Boston Scientific, BridgeBio, Bristol Myers Squibb, Corvia, Cycleron, Cytokinetics, Diastol Therapeutics, Edgewise Therapeutics, Edwards Lifesciences, Eidos, eMyosound, Ensho, Fauna Bio, Gordian, Imara, Impulse Dynamics, Intellia, Ionis, Lilly, Merck, MyoKardia, Novartis, Novo Nordisk, OrbiMed, Pfizer, Prothena, ReCor, Regeneron, Rivos, SalubrisBio, Sardocor, Shifamed, Tectonic, Tenax, Tenaya, Ulink Labs, and Ultromics. Dr Borlaug is supported by R01 HL128526, R01 HL162828, and U01 HL160226 from the National Heart, Lung, and Blood Institute, W81XWH2210245 from the U.S. Department of Defense, and the Schoen Foundation; has received research support from the NIH and the U.S. Department of Defense, as well as research grant funding from AstraZeneca, Axon, Corvia, Novo Nordisk, and Tenax Therapeutics; has served as a consultant for Actelion, Amgen, Aria, Axon Therapies, BD, Boehringer Ingelheim, Cytokinetics, Edwards Lifesciences, Lilly, Imbria, Janssen, Merck, Novo Nordisk, NGM, NXT, and VADovations; and is named inventor (U.S. Patent number 10,307,179) for the tools and approach for a minimally invasive pericardial modification procedure to treat heart failure. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Barry A. Borlaug, Mayo Clinic and Foundation, 200 First Street SW, Rochester, Minnesota 55905, USA. E-mail: borlaug.barry@mayo.edu.

PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE:

Longer duration of HFpEF/HFmrEF diagnosis is associated with more advanced structural/functional remodeling (including atrial myopathy and right heart dysfunction) and a higher risk profile. Earlier identification and staging of HFpEF/HFmrEF may help initiate therapies before progression to more advanced remodeling, while treatment response to atrial shunt therapy was not meaningfully modified by HF duration.

TRANSLATIONAL OUTLOOK: Prospective studies are needed to better define disease onset and validate HF duration as a practical staging marker linked to longitudinal remodeling, hemodynamics, and outcomes. Trials should test whether earlier-stage phenotypes derive greater benefit from targeted pharmacologic and device based treatment strategies.

REFERENCES

- Borlaug BA, Sharma K, Shah SJ, Ho JE. Heart failure with preserved ejection fraction: JACC Scientific Statement. *J Am Coll Cardiol*. 2023;81:1810-1834.
- Selvaraj S, Myhre PL, Vaduganathan M, et al. Application of diagnostic algorithms for heart failure with preserved ejection fraction to the community. *JACC Heart Fail*. 2020;8:640-653.
- Patel RB, Shah SJ. Therapeutic targeting of left atrial myopathy in atrial fibrillation and heart failure with preserved ejection fraction. *JAMA Cardiol*. 2020;5:497-499.
- Patel RB, Lam CSP, Svedlund S, et al. Disproportionate left atrial myopathy in heart failure with preserved ejection fraction among participants of the PROMIS-HFpEF study. *Sci Rep*. 2021;11:4885.
- Omote K, Sorimachi H, Obokata M, et al. Bialtrial myopathy in heart failure with preserved ejection fraction. *Eur J Heart Fail*. 2024;26(2):288-298. <https://doi.org/10.1002/ehfj.3104>
- Gorter TM, Obokata M, Reddy YNV, Melenovsky V, Borlaug BA. Exercise unmasks distinct pathophysiologic features in heart failure with preserved ejection fraction and pulmonary vascular disease. *Eur Heart J*. 2018;39:2825-2835.
- Omote K, Sorimachi H, Obokata M, et al. Pulmonary vascular disease in pulmonary hypertension due to left heart disease: pathophysiologic implications. *Eur Heart J*. 2022;43:3417-3431.
- Obokata M, Reddy YNV, Melenovsky V, Pislaru S, Borlaug BA. Deterioration in right ventricular structure and function over time in patients with heart failure and preserved ejection fraction. *Eur Heart J*. 2019;40:689-697.
- Melenovsky V, Hwang S-J, Lin G, Redfield MM, Borlaug BA. Right heart dysfunction in heart failure with preserved ejection fraction. *Eur Heart J*. 2014;35:3452-3462.
- Senni M, Caravita S, Paulus WJ. Do existing definitions identify subgroup phenotypes or reflect the natural history of heart failure with preserved ejection fraction? *Circulation*. 2019;140:366-369.
- Sorimachi H, Verbrugge FH, Omote K, et al. Longitudinal evolution of cardiac dysfunction in heart failure and preserved ejection fraction with normal natriuretic peptide levels. *Circulation*. 2022;146:500-502.
- Shah SJ, Borlaug BA, Chung ES, et al. Atrial shunt device for heart failure with preserved and mildly reduced ejection fraction (REDUCE LAP-HF II): a randomised, multicentre, blinded, sham-controlled trial. *Lancet*. 2022;399:1130-1140.
- Borlaug BA, Blair J, Bergmann MW, et al. Latent pulmonary vascular disease may alter the response to therapeutic atrial shunt device in heart failure. *Circulation*. 2022;145:1592-1604.
- Berry N, Mauri L, Feldman T, et al. Transcatheter interatrial shunt device for the treatment of heart failure: rationale and design of the pivotal randomized trial to REDUCE Elevated Left Atrial Pressure in Patients with Heart Failure II (REDUCE LAP-HF II). *Am Heart J*. 2020;226:222-231.
- Lang RM, Badano LP, Mor-Avi V, et al. Recommendations for cardiac chamber quantification by echocardiography in adults: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *J Am Soc Echocardiogr*. 2015;28:1-39.e14.
- Rudski LG, Lai WW, Afilalo J, et al. Guidelines for the echocardiographic assessment of the right heart in adults: a report from the American Society of Echocardiography endorsed by the European Association of Cardiovascular Imaging. *J Am Soc Echocardiogr*. 2010;23:685-713. quiz 786-8.
- Nagueh SF, Smiseth OA, Appleton CP, et al. Recommendations for the evaluation of left ventricular diastolic function by echocardiography: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *J Am Soc Echocardiogr*. 2016;29:277-314.
- Badano LP, Kolias TJ, Muraru D, et al. Standardization of left atrial, right ventricular, and right atrial deformation imaging using two-dimensional speckle tracking echocardiography: a consensus document of the EACVI/ASE/Industry Task Force to standardize deformation imaging. *Eur Heart J Cardiovasc Imaging*. 2018;19:591-600.
- Reddy YNV, Obokata M, Egbe A, et al. Left atrial strain and compliance in the diagnostic evaluation of heart failure with preserved ejection fraction. *Eur J Heart Fail*. 2019;21:891-900.
- Sugimoto T, Robinet S, Dulgheru R, et al. Echocardiographic reference ranges for normal left atrial function parameters: results from the

EACVI NORRE study. *Eur Heart J Cardiovasc Imaging*. 2018;19:630–638.

21. Harada T, Sorimachi H, Obokata M, et al. Echocardiographic diastolic function grading in HFpEF: testing the updated 2025 ASE criteria. *J Am Coll Cardiol*. 2026;87:1261–1275.

22. Pocock SJ, Ariti CA, Collier TJ, Wang D. The win ratio: a new approach to the analysis of composite endpoints in clinical trials based on clinical priorities. *Eur Heart J*. 2012;33:176–182.

23. Melenovsky V, Hwang S-J, Redfield MM, Zakeri R, Lin G, Borlaug BA. Left atrial remodeling and function in advanced heart failure with preserved or reduced ejection fraction. *Circ Heart Fail*. 2015;8:295–303.

24. Freed BH, Daruwalla V, Cheng JY, et al. Prognostic utility and clinical significance of cardiac mechanics in heart failure with preserved ejection fraction: importance of left atrial strain: importance of left atrial strain. *Circ Cardiovasc Imaging*. 2016;9. <https://doi.org/10.1161/CIRCIMAGING.115.003754e003754>

25. Shah AM, Claggett B, Sweitzer NK, et al. Prognostic importance of impaired systolic function in heart failure with preserved ejection fraction and the impact of spironolactone. *Circulation*. 2015;132:402–414.

26. Sorimachi H, Omote K, Borlaug BA. Clinical phenogroups in heart failure with preserved ejection fraction. *Heart Fail Clin*. 2021;17:483–498.

27. Larson K, Omar M, Sorimachi H, et al. Clinical phenogroup diversity and multiplicity: impact on mechanisms of exercise intolerance in heart failure with preserved ejection fraction. *Eur J Heart Fail*. 2024;26(3):564–577.

28. Tada A, Naser JA, Doi S, et al. Progression from exercise-induced to resting left atrial hypertension in HFpEF: impact of reduced atrio-pulmonary compliance. *JACC Heart Fail*. Published online February 10, 2026. <https://doi.org/10.1016/j.jchf.2026.102947>

29. Pfeffer MA, Shah AM, Borlaug BA. Heart failure with preserved ejection fraction in perspective. *Circ Res*. 2019;124:1598–1617.

30. Reddy YNV, Obokata M, Verbrugge FH, Lin G, Borlaug BA. Atrial dysfunction in patients with heart failure with preserved ejection fraction and atrial fibrillation. *J Am Coll Cardiol*. 2020;76:1051–1064.

31. Litwin SE, Komtebedde J, Hu M, et al. Exercise-induced left atrial hypertension in heart failure with preserved ejection fraction. *JACC Heart Fail*. 2023;11:1103–1117.

32. Gustafsson F, Petrie MC, Komtebedde J, et al. 2-Year outcomes of an atrial shunt device in HFpEF/HFmrEF: results from REDUCE LAP-HF II. *JACC Heart Fail*. 2024;12:1425–1438.

KEY WORDS cardiac remodeling, disease duration, exercise hemodynamics, heart failure with mildly reduced ejection fraction, heart failure with preserved ejection fraction

APPENDIX For supplemental tables, please see the online version of this paper.